Structural Design

Tree Fort

Report Example

Proj. 0003

Contents

1.1-1 | Summary

The design loads are based on the standards of the California Building Code (CBC) outlined below. Standard math symbols are profided for reference.

Table 1.1: California Building Standards

Standard	Organization	Description & Application
CBC Chapter 23	California Building Standards Commission	Governs the structural use of wood, including lumber, glulam, timber poles, and diaphragms.
CRC Chapters 5-8	California Building Standards Commission	Covers prescriptive construction requirements for one- and two-family light-frame wood structures.
AWC NDS	American Wood Council	Foundational national standard for ASD and LRFD design of timber structures.
AWC WFCM	American Wood Council	Design criteria for conventional light- frame wood construction to meet wind and seismic loads.
ANSI/APA PRG 320	APA â€“ The Engineered Wood Association	Standard for Performance-Rated CLT, essential for tall wood/mass timber buildings.
ICC 400	International Code Council	Design and construction standard for log structures.
ASTM Standards	ASTM International	Includes testing and evaluation methods such as ASTM D3957 (structural logs) and ASTM D7672 (rim boards).
AWPA U1 / M4	American Wood Protection Association	Standards for preservative treatment and use of wood poles, piles, and lumber in permanent installations.

Table 1.2: Math Abbreviations

Abbreviation	Definition
D	dead load
L	live load
D_m	module dead load
E	earthquake load
F_a	acceleration site coefficient
F_v	velocity site coefficient
F_N	normal wind force
GC_{Ms}	net moment static coefficient
GC_{Md}	net moment dynamic coefficient
GC_M	net moment coefficient
GC_P	net pressure coefficient
GC_{Ps}	net static pressure coefficient
GC_{Pd}	net dynamic pressure coefficient

Abbreviation	Definition
k_1	hazard coefficient
k_2	terrain and structure coefficient
k_3	topography coefficient
K_{zt}	topographic Factor
K_z	velocity pressure exposure coefficient
MRI	mean return interval
p_d	net design wind pressure on module - Pa
$SDOF$	single degree of freedom
S_s	short period mapped acceleration
S_DS	site design response acceleration
S_1	1 second period mapped acceleration
S_MS	short period parameter
S_{M1}	1 second period parameter
T	fundamental period of structure
M_{tor}	wind moment about panel center
T_0	short period spectral cap
T_S	long period spectral cap
V_b	basic wind speed
V_B	seismic design base shear
W	wind load / seismic weight of structure
F_b	NDS - Reference bending design value
F_t	NDS - Reference tension design value
F_v	NDS - Reference shear design value
F_{Cperp}	NDS - Reference compression perpendicular to grain
F_c	NDS - Reference compression parallel to grain
E	NDS - Modulus of elasticity
E_{min}	NDS - Reference modulus for stability and stiffness
G	NDS - Specific gravity
C_D	NDS - Load duration factor
C_M	NDS - Wet service factor
C_F	NDS - Size factor
C_{fu}	NDS - Flat use factor
C_i	NDS - Incising factor
C_r	NDS - Repetitive member factor
C_t	NDS - Temperature factor
C_c	NDS - Curvature factor

Abbreviation	**Definition**
C_I	NDS - Column stability factor
C_L	NDS - Beam stability factor
C_p	NDS - Column stability factor
C_b	NDS - Bearing area factor
C_T	NDS - Buckling stiffness factor
C_s	NDS - Slenderness factor
C_T	NDS - Buckling stiffness factor
F'_b	NDS - Adjusted bending design value
F'_t	NDS - Adjusted tension design value
F'_v	NDS - Adjusted shear design value
F'_{Cperp}	NDS - Adjusted compression perpendicular
F'_c	NDS - Adjusted compression parallel to grain
E'	NDS - Adjusted modulus of elasticity
E'_{min}	NDS - Adjusted modulus ofor stability and stiffness

1.2-1 | Load Combinations and Geometry

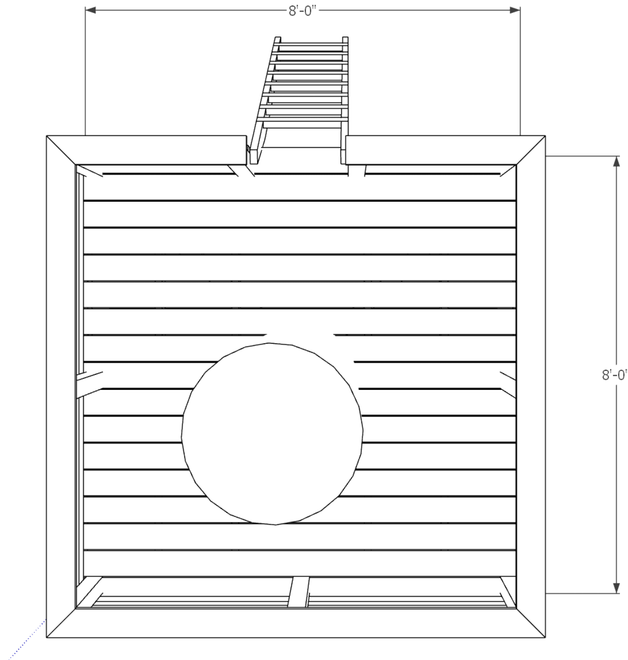


Fig. 1.1 - Tree Fort Plan

Table 1.1: ASCE 7-05 Load Effects

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

1.2-2 | Unit Loads

Unit weights imported from csv file - table created by AI.

Table 2.1: Unit Weights - Doug Fir

Nominal	ActualWidth_in	ActualDepth_in	Area_in2	Volume_ft3_per_ft	Density_lb_per_ft3	Weight_lb_per_ft
2x2	1.5	1.5	2.25	0.015625	34	0.53125
2x4	1.5	3.5	5.25	0.0364583	34	1.23958
2x6	1.5	5.5	8.25	0.0572917	34	1.94875
2x8	1.5	7.25	10.875	0.0755208	34	2.56771
2x10	1.5	9.25	13.875	0.0963542	34	3.275
2x12	1.5	11.25	16.875	0.117188	34	3.98438

Nominal	ActualWidth_in	ActualDepth_in	Area_in2	Volume_ft3_per_ft	Density_lb_per_ft3	Weight_lb_per_ft
4x4	3.5	3.5	12.25	0.0850694	34	2.89236

Variables assigned by inline definitions.

Table 2.2: Member Nominal Loads and Properties

variable	value	[value]	description
D_1	2.00 lb_ft	0.03 kN_m	2x6 planks DL
D_2	2.60 lb_ft	0.04 kN_m	2x8 joists DL
D_3	2.90 lb_ft	0.04 kN_m	4x4 posts and struts
E_1	29000.00 k_si	199947.96 MPA	modulus of elasticity
LL_1	40.00 p_sf	1.92 kPA	ASCE7-05 floor LL
HL_1	20.00 p_sf	0.96 kPA	ASCE7-05 HL

1.3-1 | Report Summary

This report covers the structural design of a tree fort in Novato, California, following the California Building Code (CBC). The fort is supported by a mature tree with a 24-inch diameter trunk.

The report illustrates the use of tags, commands and scripts including:

- a rivt-report.py script that assembles the report.
- shell commands that run an **<** external program **>** .
- external urls and section links between documents.
- various forms of math symbols.
- external function importing and processing.
- the use of AI in preparing

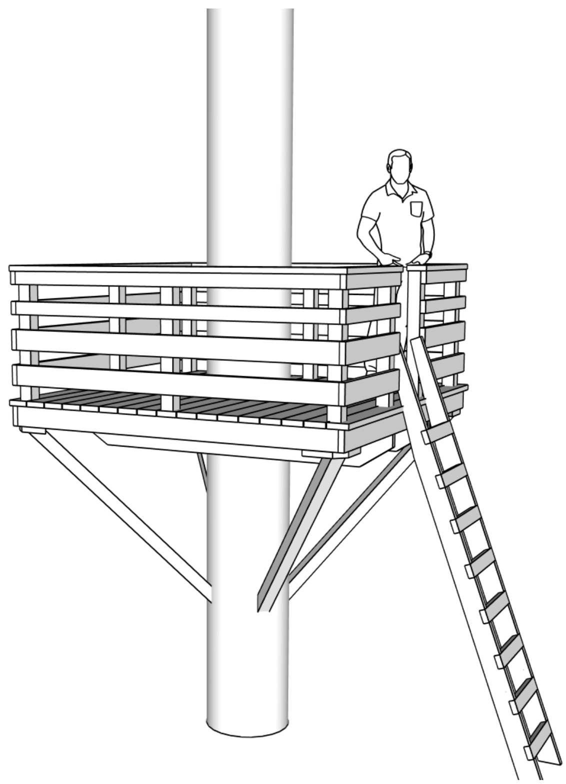


Fig. 1.1 - Tree Fort

1.3-2 | Design Sections

The design report is organized into the following sections:

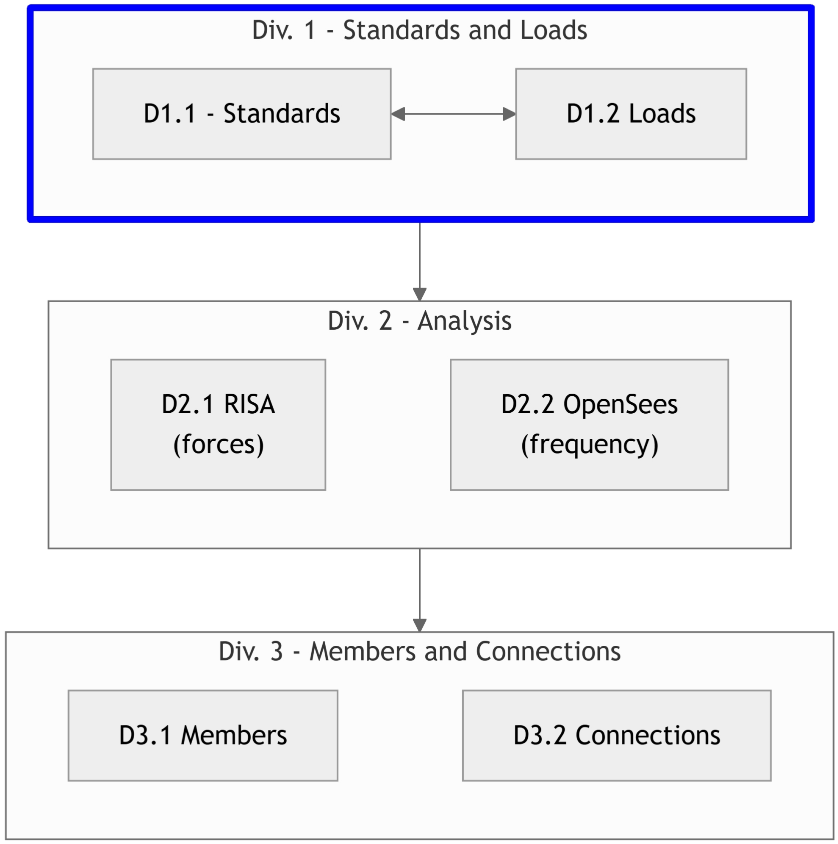


Fig. 2.1 - Report Flow Chart

1.3-3 | Drawing Symbols

Table 3.1: Drawing Abbreviations

Abbreviation	Definition
ASD	Allowable Stress Design
ACI	American Concrete Institute
AISC	American Institute of Steel Construction
AISI	American Iron and Steel Institute
ASTM	American Society for Testing and Materials
AWS	American Welding Society
AB	Anchor Bolt
BDRY	Boundry
CBC	Califiornia Building Code
CRC	Califiornia Residential Code
CIP	Cast-In-Place
CLR	Clear

Abbreviation	Definition
CONC	Concrete
CMU	Concrete Masonry Unit
CRSI	Concrete Reinforcing Steel Institute
CONST JT	Construction Joint
CONT	Continuous
CJ	Control Joint
D-C	Demand-Capacity (ratio)
DIA	Diameter
DIM	Dimension
EA	Each
EF	Each Face
EJ	Expansion Joint
ES	Each Side
EW	Each Way
EXP Bolt	Expansion Bolt
EXP JT	Expansion Joint
FTG	Footing
FND	Foundation
GALV	Galvanized
GA	Gauge
GR	Grade
HT	Height
IN	Inch
ID	Inside Diameter
ICBO	International Conference of Building Officials
K	Kip (1000 Pounds)
LWC	Light Weight Concrete
LRFD	Load and Resistance Factor Design
NWC	Normal Weight Concrete
NIC	Not in Contract
OC	On Center
OD	Outside Diameter
OPNG	Opening
PVC	Polyvinyl Chloride
PSF	Pounds per Square Foot
PSI	Pounds per Square Inch

Abbreviation	Definition
R	Radius
REINF	Reinforced
SIM	Similar
SOG	Slab on Grade
SL	Splice Length
SQ	Square
STD	Standard
SDI	Steel Deck Institute
SF	Step Footing or Square Foot
SYM	Symmetrical
THK	Thick or Thickness
T&B	Top and Bottom
T&G	Tongue and Groove
TOC	Top of Concrete
TOF	Top of Foundation
TOS	Top of Steel
TOW	Top of Wall
TYP	Typical
UNO	Unless Noted Otherwise
WWF	Welded Wire Fabric
W/	With
WP	Working Point

2.1-1 | Analysis Programs

This division includes RISA and OpenSees analysis and illustrates methods for running and importing outputs from external programs.

- RISA-3D for determining connections forces
- OpenSees for determing the period of the tree fort system.

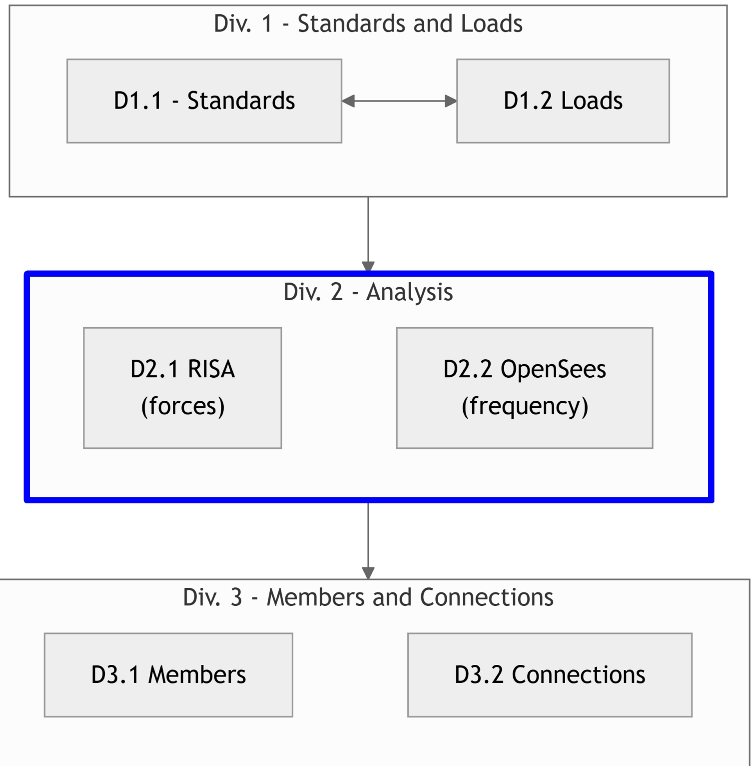


Fig. 1.1 - Report Flow Chart

2.2-1 | OpenSees Analysis

```
This section analyzes the period of the tree + tree fort system.
The OpenSees 2 DOF model is shown schematically.

        y
        ^
        |
    m2  o  Node 3 (Branches)
        |
    k2  |   Spring 2 (upper trunk)
        |
    m1  o  Node 2 (Tree Fort)
        |
    k1  |   Spring 1 (lower trunk)
        |
        o  Node 1 (Fixed Base)
    -----------
    (Ground)
```

2.2-2 | OpenSees values

variable	value	[value]	description
mass1	1.5	1.5	mass of tree fort, kN/g
mass2	3.5	3.5	mass of branches, kN/g
trk1	1100	1100	lower tree trunk stiffness, kN/cm
trk2	2100	2100	upper tree trunk stiffness, kN/cm

2.2-3 | OpenSees Model Input

This is the Python code for a 2 DOF model of the tree fort with substituted values from the rivt file. The OpenSees file with substitutions is written to _rvstor/data.

```python
# opensees model for a 2-DOF system

import openseespy.opensees as ops
import numpy as np
import math
import matplotlib.pyplot as plt
import opsvis as opsv
# ---------------------------------------------------------------------
# 1. Model Initialization
# ---------------------------------------------------------------------
ops.wipe()
ops.model("BasicBuilder", "-ndm", 2, "-ndf", 2)

# ---------------------------------------------------------------------
```

```
# 2. Define Parameters
# ---------------------------------------------------------------------
# Mass values (e.g., in tons)
m1 = 1.5
m2 = 3.5

# Stiffness values (e.g., in kN/m)
k1 = 1100.0
k2 = 2100.0

# ---------------------------------------------------------------------
# 3. Create Nodes
# ---------------------------------------------------------------------
# Base node (fixed)
ops.node(1, 0.0, 0.0)

# Node 1
ops.node(2, 0.0, 2.0)

# Node 2
ops.node(3, 0.0, 4.0)

# Fix the base node in both X and Y directions
ops.fix(1, 1, 1)
# Restrain vertical (Y) displacement and rotation at mass nodes
# to represent a pure 2D shear/lateral lollipop system
ops.fix(2, 0, 1)  # Free in X, Fixed in Y
ops.fix(3, 0, 1)  # Free in X, Fixed in Y
# ---------------------------------------------------------------------
# 4. Assign Masses
# ---------------------------------------------------------------------

ops.mass(2, m1, 0.0)
ops.mass(3, m2, 0.0)

# ---------------------------------------------------------------------
# 5. Define Elements
# ---------------------------------------------------------------------
# Use elastic truss elements to represent the lateral springs
# Assign high axial stiffness (E * A) and 0.0 length
EA = 1e8

# Spring 1 between Node 1 and Node 2
ops.uniaxialMaterial("Elastic", 1, 1100.0)
ops.element("Truss", 1, 1, 2, EA, 1)

# Spring 2 between Node 2 and Node 3
ops.uniaxialMaterial("Elastic", 2, 2100.0)
ops.element("Truss", 2, 2, 3, EA, 2)

# ---------------------------------------------------------------------
# 6. Eigenvalue Analysis & Results Output
# ---------------------------------------------------------------------
num_modes = 1
```

```
eigenvalues = ops.eigen(num_modes)

periods = []
frequencies = []
outL = []
for i in range(num_modes):
    lamb = eigenvalues[i]
    omega = math.sqrt(lamb)
    freq = omega / (2 * math.pi)
    period = 2 * math.pi / omega
    periods.append(period)
    frequencies.append(freq)
    resS = f'Mode {i + 1}: Period = {period:.4f} s | Frequency = {freq:.4f} Hz'
    outL.append(resS)
outS = chr(10).join(item for item in outL)
with open("output.txt", 'w') as f1:
    f1.write(outS)
print("text output written")
# ---------------------------------------------------------------------
# 7. Model Visualization
# ---------------------------------------------------------------------

# Plot the defined model to check geometry (this is an axes mdoel)
mod1 = opsv.plot_model()
fig1 = mod1.get_figure()
plt.title("2-DOF Lollipop Model Geometry")
fig1.savefig("figure1.png", dpi=200)
print("figure 1 written")
# interactive display
# plt.show(block=False)

# Plot the mode shape for the first mode (this is a figure)
fig2 = opsv.plot_mode_shape(1, 2.0)  # Scaling factor of 2.0 for visibility
plt.title("Mode Shape 1")
plt.savefig("figure2.png", dpi=200)
print("figure 2 written")
# plt.show()
```

2.2-4 | OpenSees Shell

The model is run in a virtual environment with Python 3.12 and OpenSeesPyinstalled. The model calculates the natural frequencies and mode shapes ofthe system. This section runs a shell command in a virtual Python environment forPython-3.12 - the version compatible with OpenSeesPy.**File written to: /_rvstore/data/ops1.cmd**

```
cd "%USERPROFILE%"
cd py312
call scripts\activate
python opsmod1.py
```

Copied opsmod1.py from .../rivt-report/rvsrc/scripts to .../C:/Users/rhh2

Run shell command _[B]**Run “c:gitrivt-example-03-gitrivt-reportrvsrcscriptsops1.cmd”**

Copy images and text from OpenSees analysis _[B]**Copied *.png from …/rivt-report/rvsrc/scripts to …/-rvsrc-***

Copied *.txt from …/rivt-report/rvsrc/scripts to …/-rvsrc-

2.2-5 | Model Plots and Output

Analysis Output

```
Mode 1: Period = 11.7548 s | Frequency = 0.0851 Hz
```

Model Plots

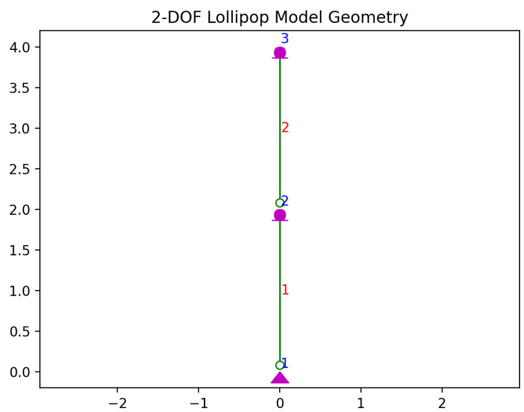


Fig. 5.1 - OPS Model

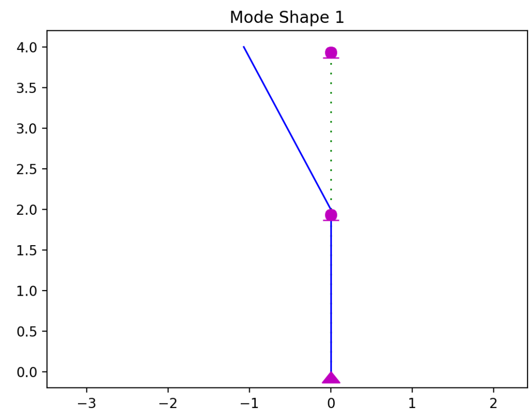


Fig. 5.2 - OPS First Mode

2.3-1 | Applied Deck and Railing forces - RISA model

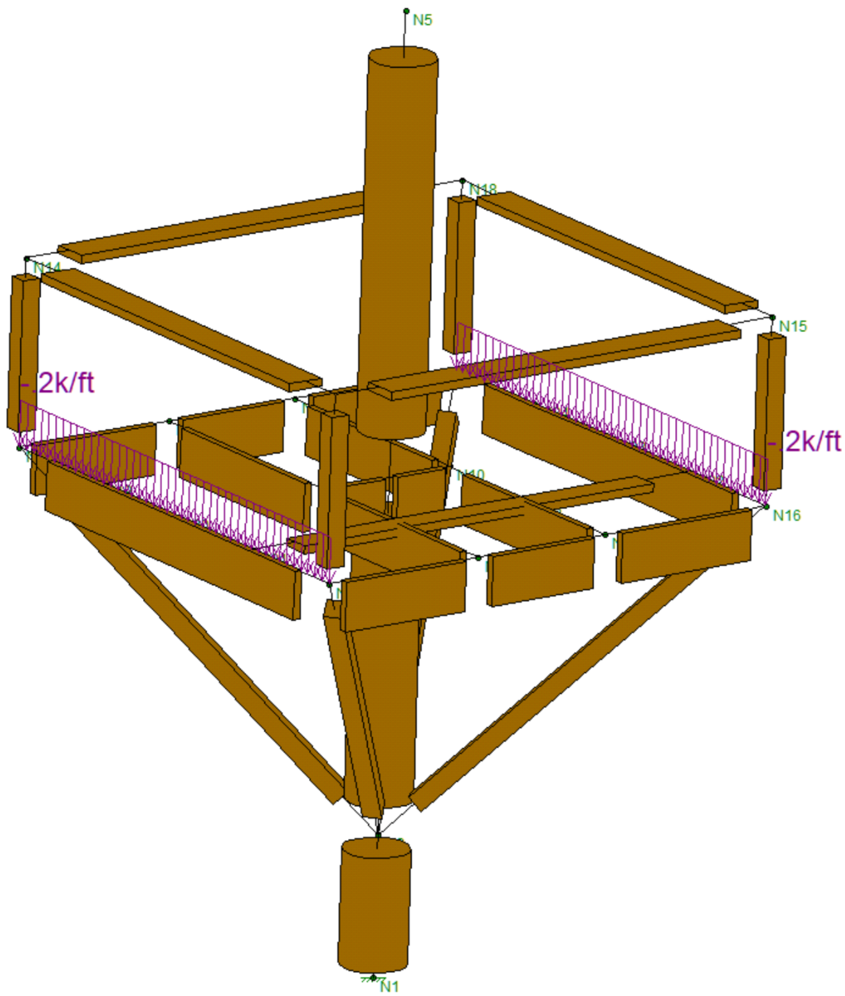


Fig. 1.1 - Risa Model

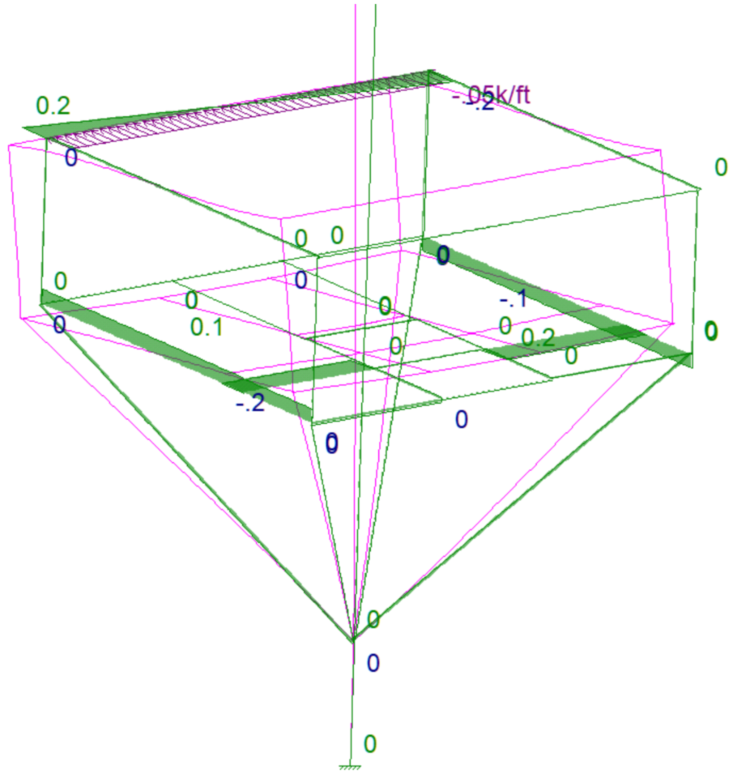


Fig. 1.2 - Rail Lateral Forces

2.3-2 | Resultant axial forces - RISA model

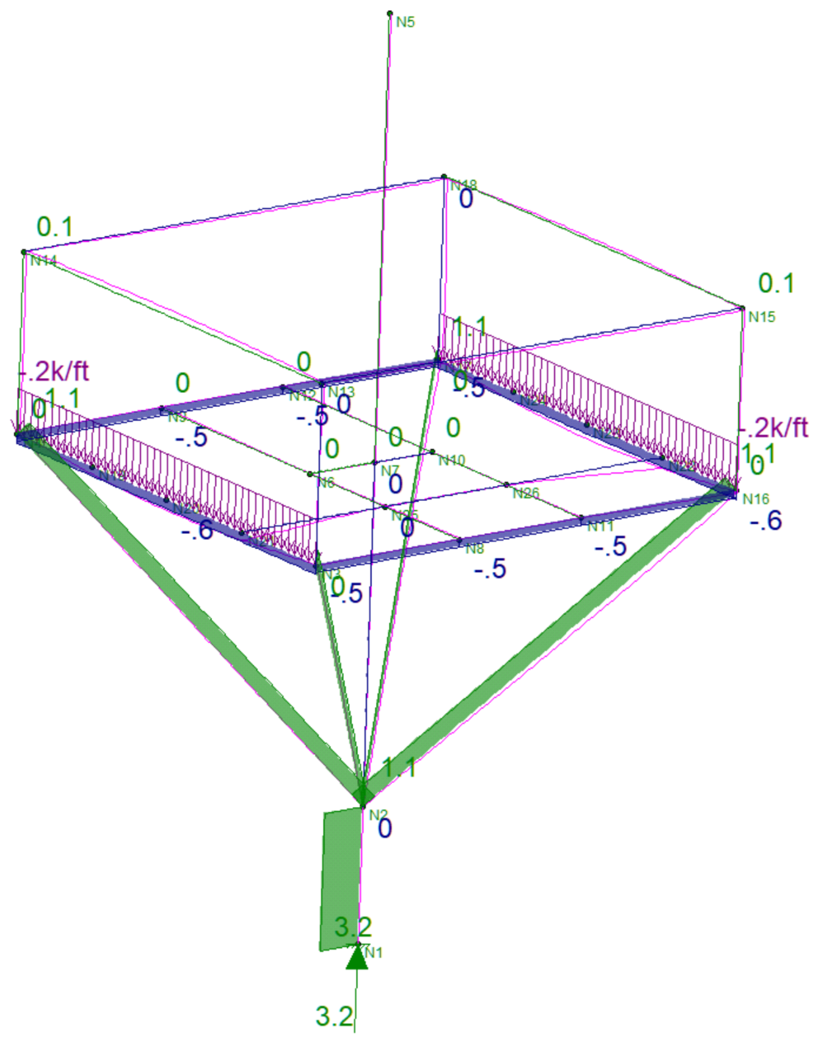


Fig. 2.1 - Strut Axial Forces

2.3-3 | Top rail shear reactions - RISA model

```
Under the California Building Code (CBC), handrails and guard railings
must resist a uniform load of 50 plf and a concentrated point load of 200
lbs, both applied horizontally to the top rail. Intermediate rails,
balusters, and infill panels must separately withstand a concentrated load
of 50 lbs.

Structural Schematic of Railing and Loads (drawn by AI)

  =======  <-- Top Rail
     |     <-- Lateral Load (P)
     |
     |
     |
     |  Height (h)
     |
     |
     |
     |
===========  <-- Fixed Support / Deck Surface
```

3.1-1 | Component Design

This division covers the design of members and connections.

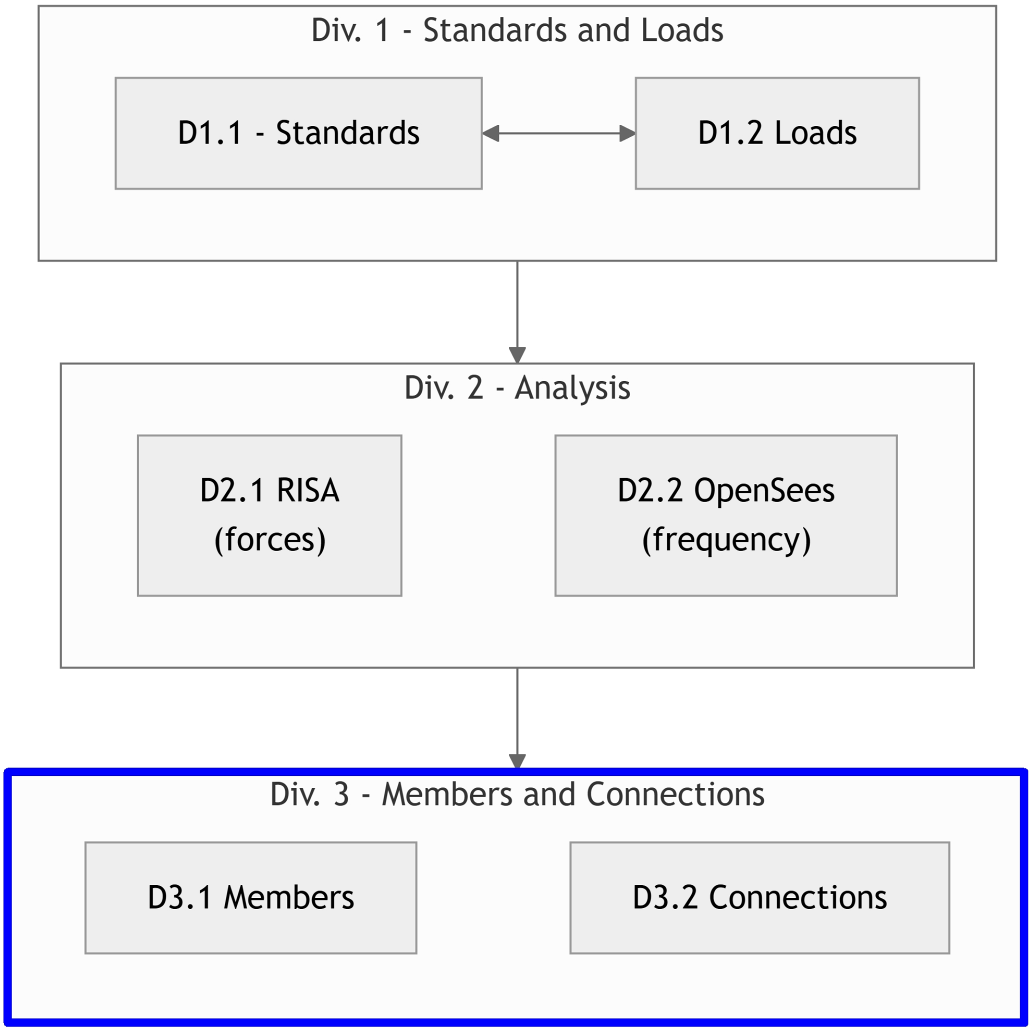


Fig. 1.1 - Report Flow Chart

3.1-2 | Geometry and Components

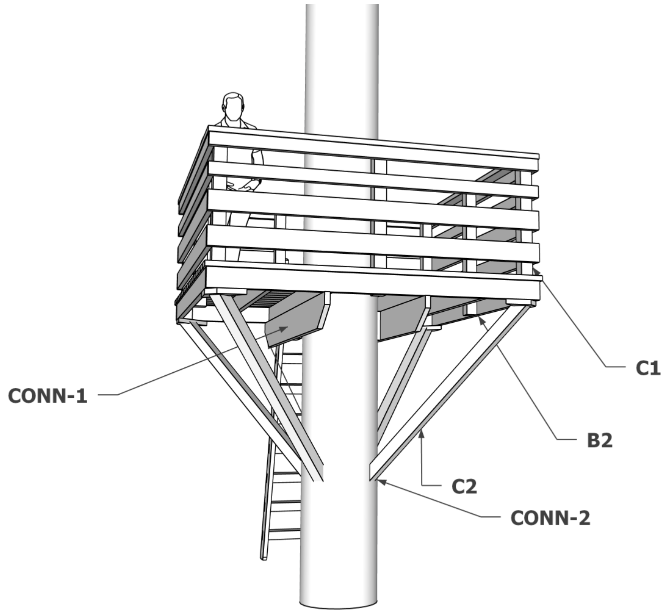


Fig. 2.1 - Component Labels

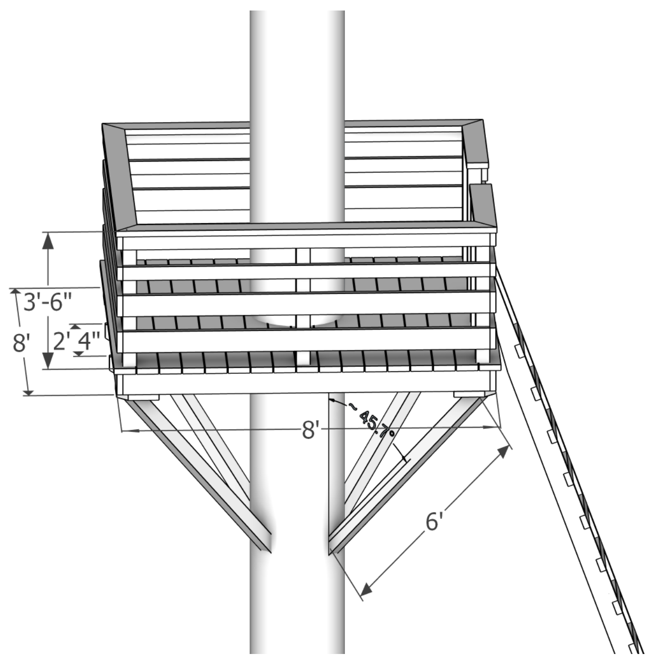
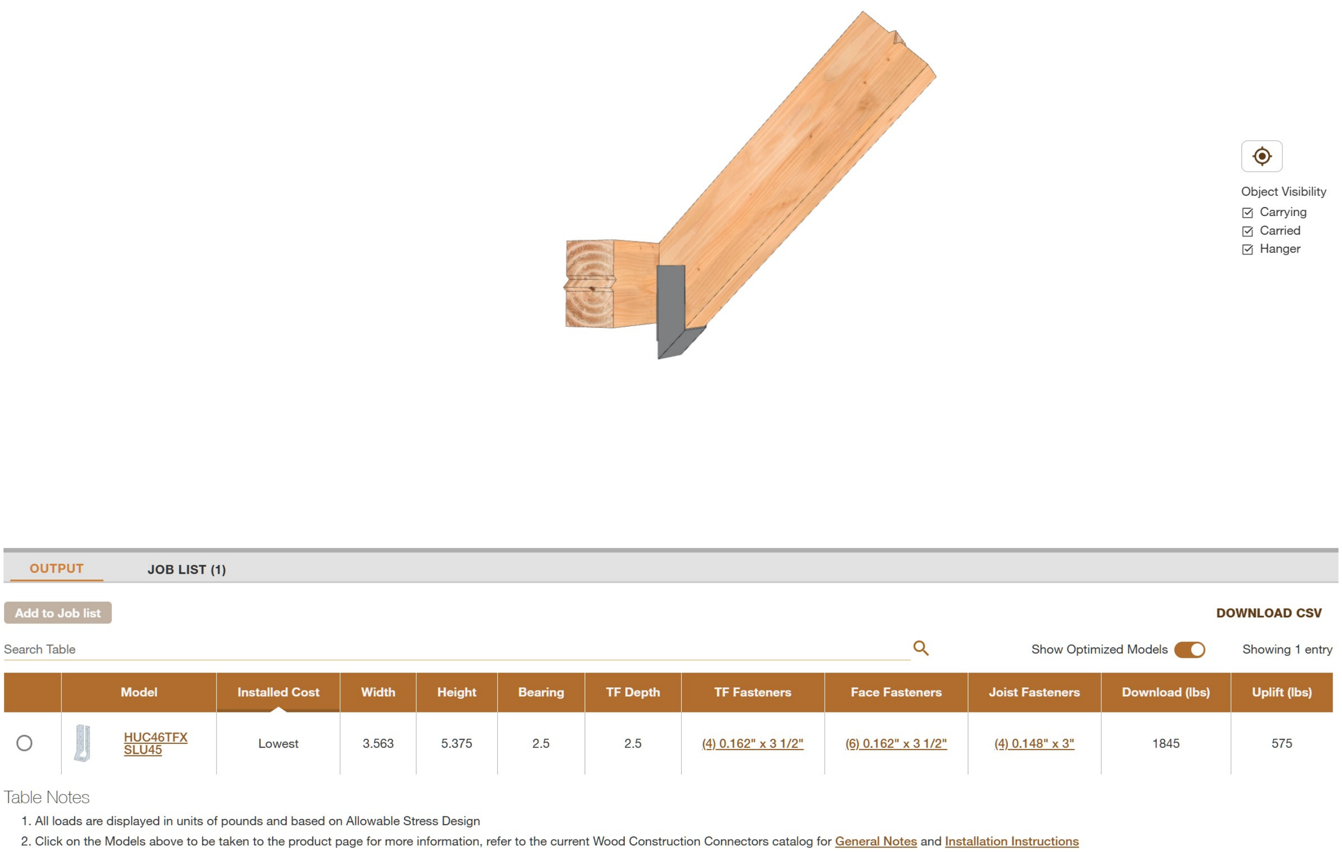


Fig. 2.2 - Dimensions

3.2-1 | Strut to Tree Connection

Use Simpson Strong Tie online selection tool.

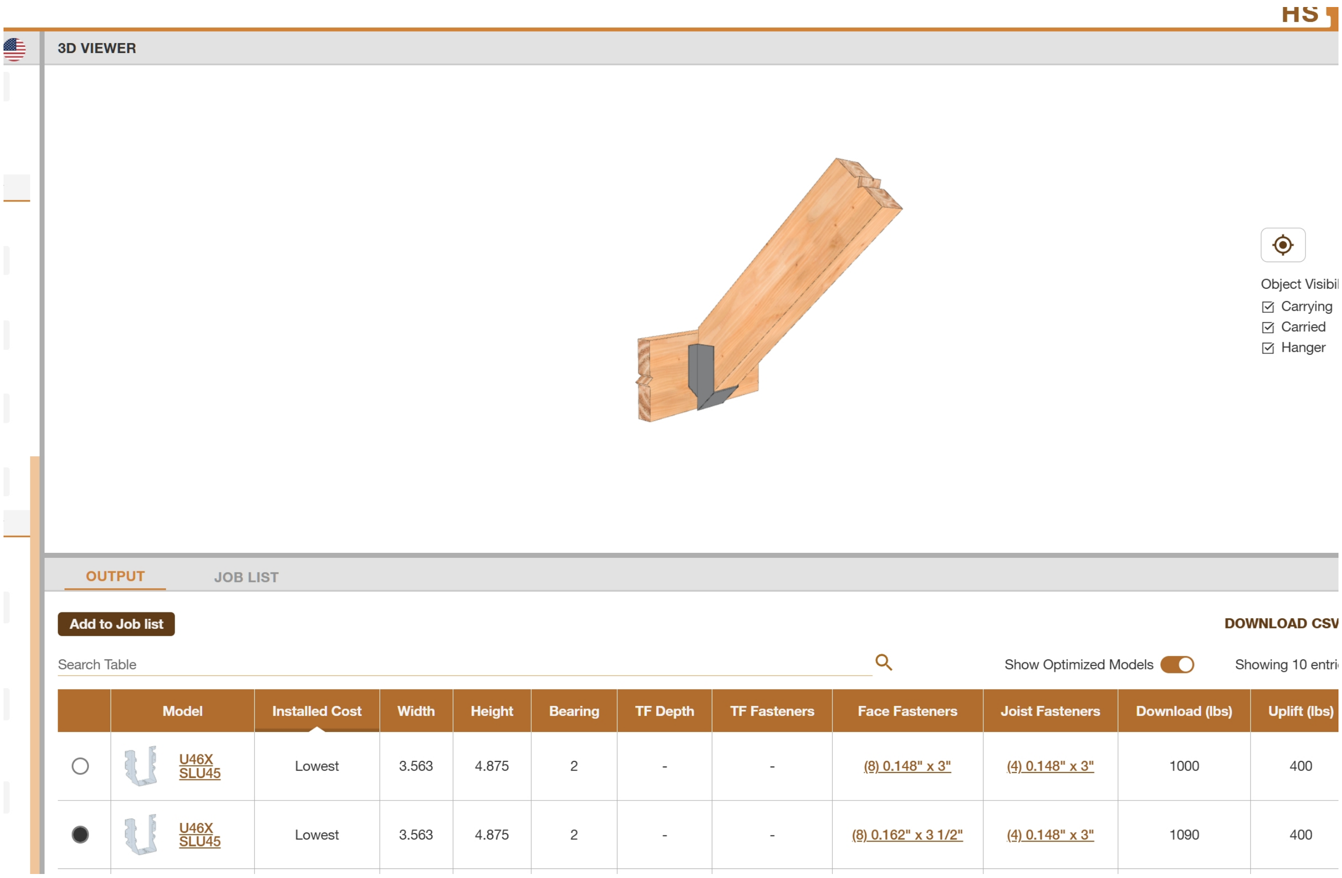


	Model	Installed Cost	Width	Height	Bearing	TF Depth	TF Fasteners	Face Fasteners	Joist Fasteners	Download (lbs)	Uplift (lbs)
	HUC46TFX SLU45	Lowest	3.563	5.375	2.5	2.5	(4) 0.162" x 3 1/2"	(6) 0.162" x 3 1/2"	(4) 0.148" x 3"	1845	575

Table Notes

1. All loads are displayed in units of pounds and based on Allowable Stress Design
2. Click on the Models above to be taken to the product page for more information, refer to the current Wood Construction Connectors catalog for General Notes and Installation Instructions

Fig. 1.1 - Screen: Option 1 | time: no time



	Model	Installed Cost	Width	Height	Bearing	TF Depth	TF Fasteners	Face Fasteners	Joist Fasteners	Download (lbs)	Uplift (lbs)
○	U46X SLU45	Lowest	3.563	4.875	2	-	-	(8) 0.148" x 3"	(4) 0.148" x 3"	1000	400
●	U46X SLU45	Lowest	3.563	4.875	2	-	-	(8) 0.162" x 3 1/2"	(4) 0.148" x 3"	1090	400

Fig. 1.2 - Screen: Option 2 | time: no time

3.2-2 | Top rail Corner

Use AWC online connection tool.

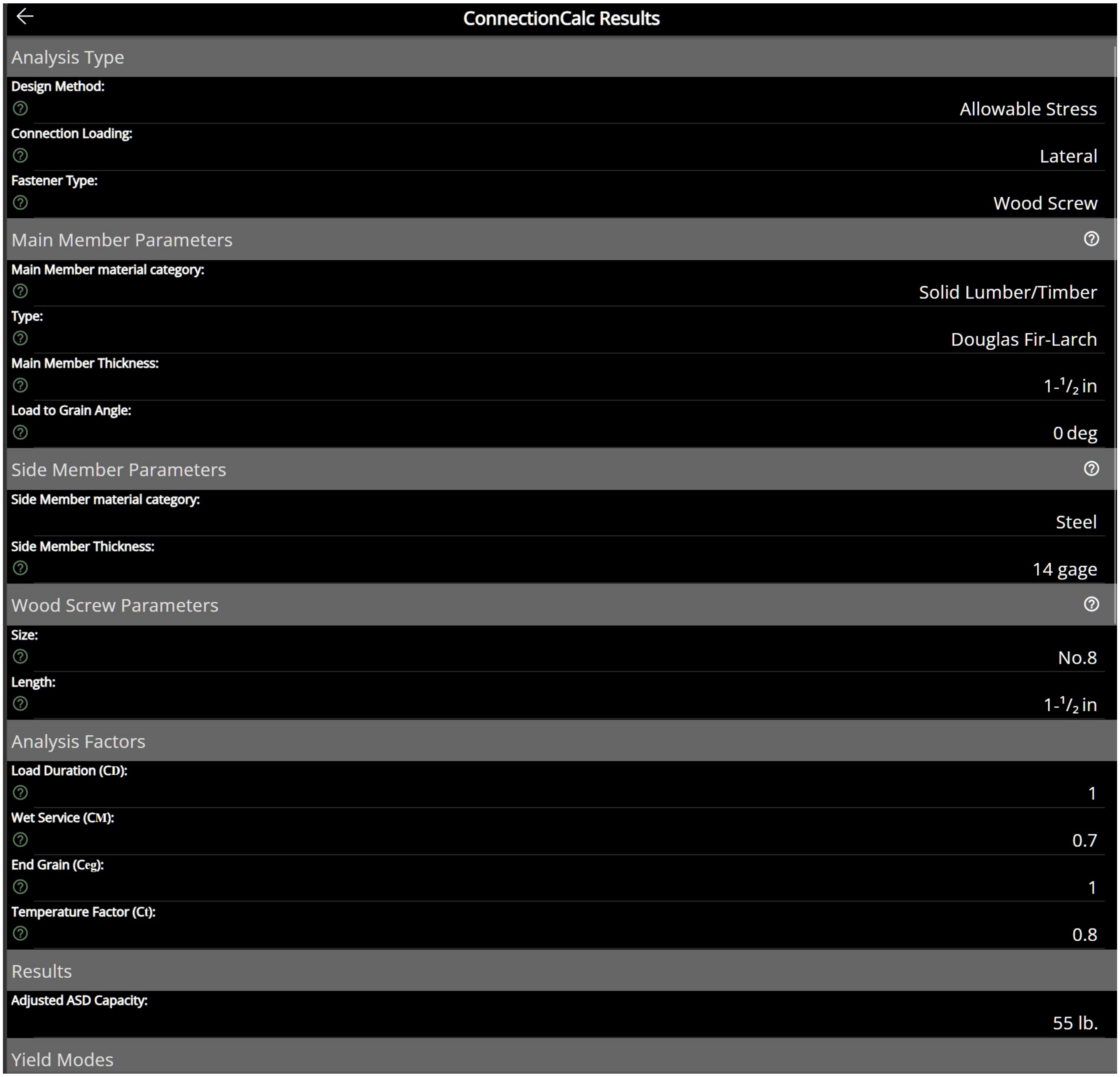


Fig. 2.1 - Screen: Top Rail - Corner Plate Input

ConnectionCalc Results

End Grain (Ceg):	1
Temperature Factor (Ct):	0.8
Results	
Adjusted ASD Capacity:	55 lb.
Yield Modes	
Z' Im:	195 lb.
Z' Is:	156 lb.
Z' II:	78 lb.
Z' IIIm:	89 lb.
Z' IIIs:	55 lb.
Z' IV:	72 lb.
Notes	
Douglas Fir- Larch Main Member Dowel Bearing Strength (psi):	4,650 psi
Side Member Assumed Steel Alloy:	ASTM A653 G33
Steel Side Member Dowel Bearing Strength (psi):	61,850 psi
Fastener Bending Yield Strength:	90,000 psi
Tip length:	2x diameter
Screw Dowel Bending Strength rounding:	nearest 50 psi
Lateral Results Rounding:	nearest 10 lb
Toe-nails in wood members:	results do not apply

Disclaimer:

While every effort has been made to insure the accuracy of the information presented, and special effort has been made to assure that the information reflects the state-of-the-art, neither the American Wood Council nor its members assume any responsibility for any particular design prepared from this Connection Calculator. Those using this Connection Calculator assume all liability from its use.

Fig. 2.2 - Screen: Top Rail - Corner Plate Capacity

Use 4-no. 8 screws = 55 lbs * 4 = Capacity 220 lbs | Demand = 200 lbs.

3.3-1 | Deck Design Properties

Import deck loads and functions.

Table 1.1: Values from rv102-loads.py (v102-2.csv)

variable	value	[value]	description
D_1	2.00 lb_ft	0.03 kN_m	2x6 planks DL
D_2	2.60 lb_ft	0.04 kN_m	2x8 joists DL
D_3	2.90 lb_ft	0.04 kN_m	4x4 posts and struts
E_1	29000.00 k_si	199947.96 MPA	modulus of elasticity
LL_1	40.00 p_sf	1.92 kPA	ASCE7-05 floor LL
HL_1	20.00 p_sf	0.96 kPA	ASCE7-05 HL

Table 1.2: Import Functions (checks.py)

Function	Docstring
nds_beam_check(** kwargs)	Check stress and deflection for a simply supported wood beam using NDS.
nds_post_check(** kwargs)	Check stress at cantilever post

3.3-2 | Deck Design Summary

Design properties as dictionary for checking function nds_beam_chk

```
Function Arguments Dictionary : beam1 (units: inch, pounds)
===========================================================================
ln_1 = 4*12.  # beam span
w_1 = 45*.5/12  # uniform linear load
b_1 = 5.5  # beam width
d_1 = 1.5  # beam depth
E_1 = 1.5*(10**6)  # modulus of elasticity
F_b = 1000 # allowable bending stress
C_D = 1.0  # load duration factor
C_M = 0.85 # wet service factor
C_F = 1.0  # size factor
C_t = 1.0  # temperature factor
C_i = 0.8  # incising factor
C_r = 1.0  # repetitive member factor
C_c = 1.0  # curvature factor
C_L = 1.0  # beam stability factor
C_b = 1.0  # bearing area factor
deflect_limit = 240.0 # max allowable deflection ln_1/deflect_limit
===========================================================================
```

Design Results

3.3-3 | Strut

Check strut D/C ratio with BeamChek 2023

BeamChek v2023 licensed to: StL Reg # 2999-68413

Tree Fort — Struts

Prepared by: Date: 6/09/26

Selection **4x 4 DF-L Select Solid Wood Column**

Conditions NDS 2018, Using values for 2x and 4x solid sawn, Dimension Lumber.

Wet Use

Data

Load	1100 #	Column Area	12.25 in²	Kf	1.00
Actual Height	8.0 ft	le d1 Effective Ht	96 in	c	0.80
Unbraced L1	8.0 ft	le d2 Effective Ht	96 in	KcE	0.30
Unbraced L2	8.0 ft	Ke Buckling Mode	1.0	FcE	683

Attributes and Values **Controlling d is 3.5 inches**

	le/d	psi	Area (in²)
Actual	27	90	12.25
Critical	50	606	1.82
Status	OK	OK	OK
Ratio	54%	15%	15%

	Fc ‖ (psi)	E (psi x mil)
Reference Values	1700	1.9
Adjusted Values	606	1.7
CF Size Factor	1.15	
Cd Duration	1.00	
Cm Wet Use	0.80	0.90
Cp Stability	0.39	

Note: A wood plate under this column must have an Fc value, perpendicular to the grain, greater than 90 psi.

Fig. 3.1 - Screenshot: Strut Check